agc162

AI vision for fruit farms.

agc162 reads leaves, flowers and fruit from photos, video and live cameras. It flags disease before it spreads and tells growers what to do next.

PITCH FOR PILOT PARTNERS · OCTOBER 2026



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A HORTUS PROJECT OF THE LASTFOMO RESEARCH HOUSE

UP TO

40%

of the world’s crop production is lost to pests and disease every year.

FAO / IPPC SCIENTIFIC REVIEW, 2021 · AGENCY ESTIMATE

Even in the five best-studied staple crops, pests and pathogens take 17–30% of yield.

SAVARY ET AL. 2019, NATURE ECOLOGY & EVOLUTION

$220B

a year: what plant disease costs the world economy.

FAO / IPPC, 2021

14%

of food is lost between harvest and retail; fruit and vegetables lose more than cereals at most stages.

FAO, STATE OF FOOD AND AGRICULTURE 2019

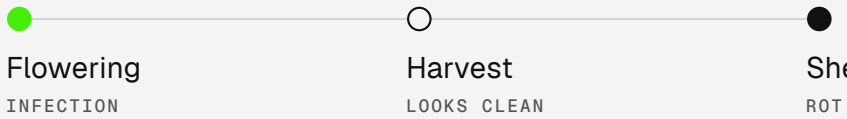
59 Mt

of mangoes, mangosteens and guavas grown in 2022; India grows about 44%.

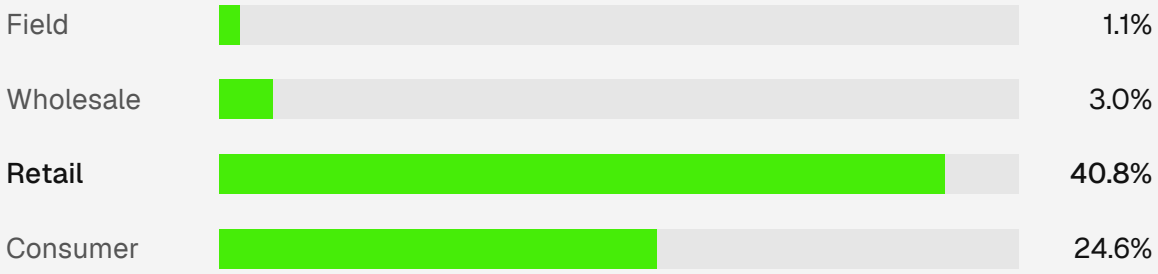
FAOSTAT VIA JIRCAS 2024

THE HIDDEN LOSS

Rot starts at flowering. It shows on the shelf.



MANGO LOST TO FUNGAL ROT, BY STAGE



ONE SURVEY, COIMBATORE, INDIA · MADRAS AGRICULTURAL JOURNAL 92: 42-48 · LIMITED EVIDENCE

Anthracnose infects flowers and young fruit in warm, wet weather, then stays latent. Fruit that looks clean at harvest can rot as it ripens.

NELSON 2008, UNIVERSITY OF HAWAI'I CTAHR PD-48

By the time anyone can see it, the fruit is sold. The moment to act is weeks earlier, at flowering.

Handling matters too: better post-harvest practice cut losses from 39.6% to 6.2% in one Sri Lankan mango chain.

HERATH ET AL. 2021 · LIMITED EVIDENCE

Today’s tools are blind or brittle.

Most

scheduled anthracnose sprays in one Philippine mango orchard fell in low-risk periods.

ESTRADA & ILAG 1990, UPLB · LIMITED

4 fewer

sprays might have been needed in one mango trial had an infection model planned them.

DODD ET AL. 1991, PLANT PATHOLOGY 40 · LIMITED

99.35% → 31.4%

accuracy of a lab-trained plant-disease model on its own test set, then on photos taken elsewhere.

MOHANTY, HUGHES & SALATHÉ 2016 · MODERATE

Grader to grader

manual mango grading is slow, inconsistent and subjective, so the price depends on who is on shift.

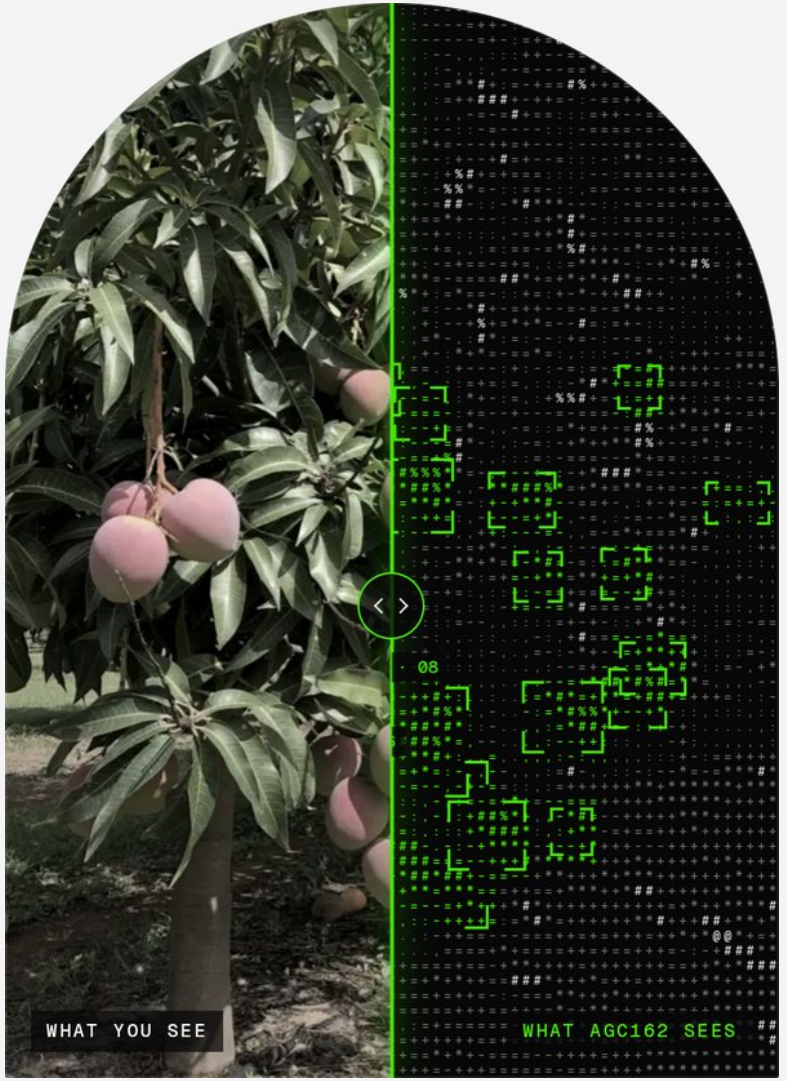
FOODS 11(21): 3483, 2022

Every tree, seen. Every week.

agc162 turns photos, video and camera feeds into a tree-by-tree record, then into a decision: what to check, when to protect and how to grade.

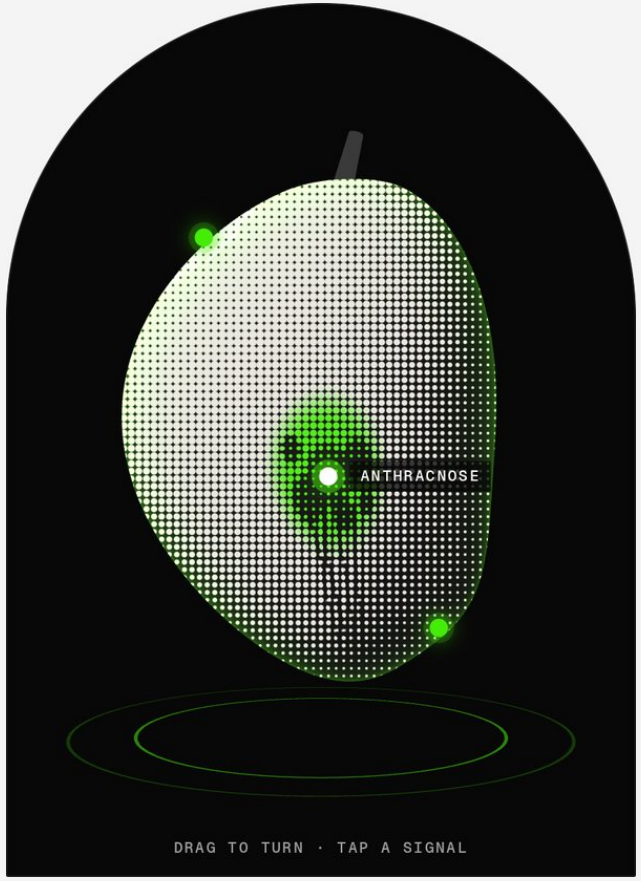
01	Capture	Phone photos, row video, drone passes or fixed cameras.
02	Detect	Fruit, flowers, lesions and pests, tree by tree.
03	Forecast	Flowering stage, leaf wetness and temperature become infection risk.
04	Treat	A window, a block and a product class, checked against the label.
05	Grade	Size and blemish at harvest and on the pack line.

BUILT ON PUBLISHED METHODS: FRUIT DETECTION ON MANGO CANOPIES (KOIRALA ET AL. 2019), PANICLE-STAGE CLASSIFICATION (KOIRALA ET AL. 2020), WEATHER-TIMED PROTECTION (DODD ET AL. 1991)



LEFT: THE PHOTO · RIGHT: WHAT AGC162 SEES

Six problems agc162 is built to flag.



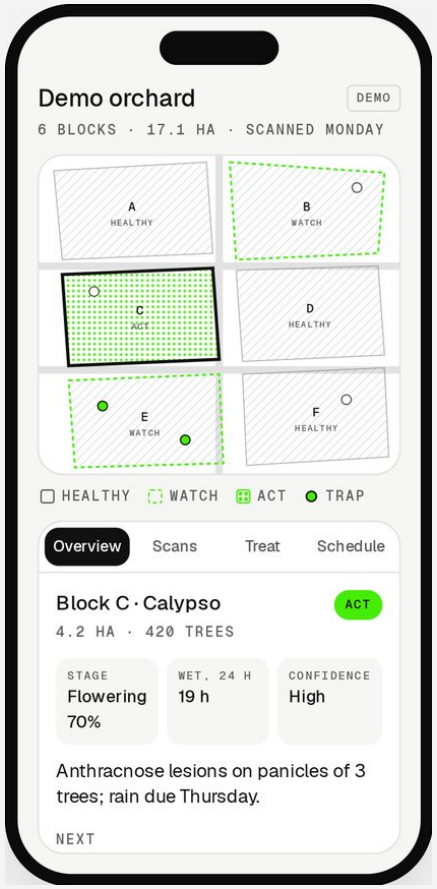
PROBLEM	LOOKS LIKE	WHEN	DO
Anthracnose Colletotrichum spp.	Dark, sunken spots; tear-stain streaks; black flecks on flowers	Infects at flowering in warm, wet weather; shows at ripening	Protect flowers before wet spells, prune for airflow, then hot-water or registered post-harvest treatment
Stem-end rot Lasiodiplodia, Neofusicoccum spp.	Dark rot spreading from the stalk end as fruit ripens	Enters via stalk and wounds; worse after stress and rough handling	Remove dead wood; careful picking, trimmed stalks, de-sapping; registered post-harvest treatment
Black spot Xanthomonas citri pv. mangiferaeindicae	Raised black spots with greasy edges that crack; angular leaf spots	Warm, humid weather with wind-driven rain	Windbreaks; copper as registered, with mancozeb instead during flowering
Powdery mildew Pseudoidium anacardii	White powder on flowers and young fruit; flowers drop	Flowering; cool nights, 10–31 °C, 60–90% humidity	Protective sulphur as panicles reach 8–10 cm; prune out infected panicles
Sooty mould Capnodium, Meliola spp.	A black film that rubs off leaves and fruit	Grows on honeydew from scale, mealybugs and hoppers	Control the sap-sucking insects that feed it
Fruit fly Bactrocera spp.	Puncture marks, weeping sap; larvae in the flesh	As fruit matures and colours	Traps and the area programme; disinfestation such as hot water for export

GENERAL GUIDANCE. USE ONLY PRODUCTS REGISTERED FOR MANGO WHERE YOU FARM, AT LABEL RATES AND WITHHOLDING PERIODS · SOURCES ON PAGE 13

One app, block to packhouse.

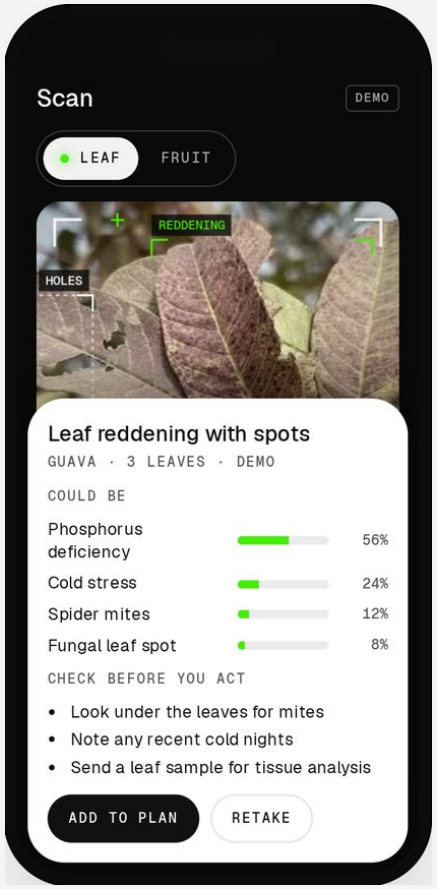
Scans roll up into an orchard map. A scan returns likely causes and what to check. The plan turns flowering stage, leaf wetness and temperature into a spray window.

DEMO SCREENS · DEMO DATA



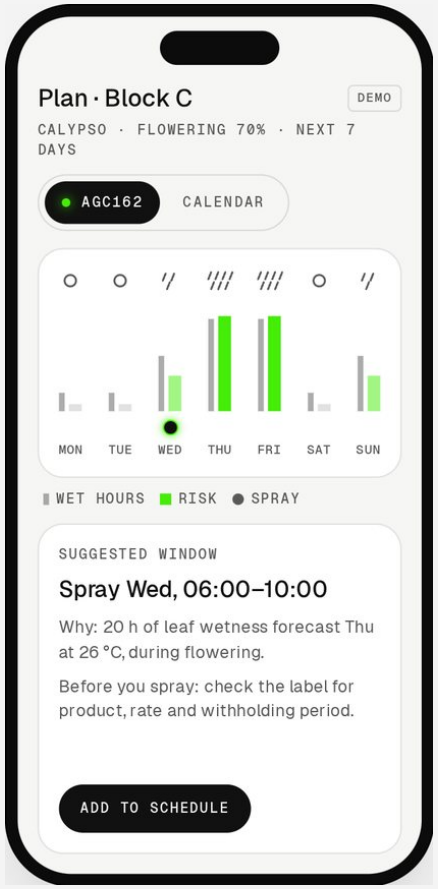
ORCHARD MAP

Every block with its status, confidence and next step.



SCAN

Likely causes with odds, and what to check before spraying.



PLAN

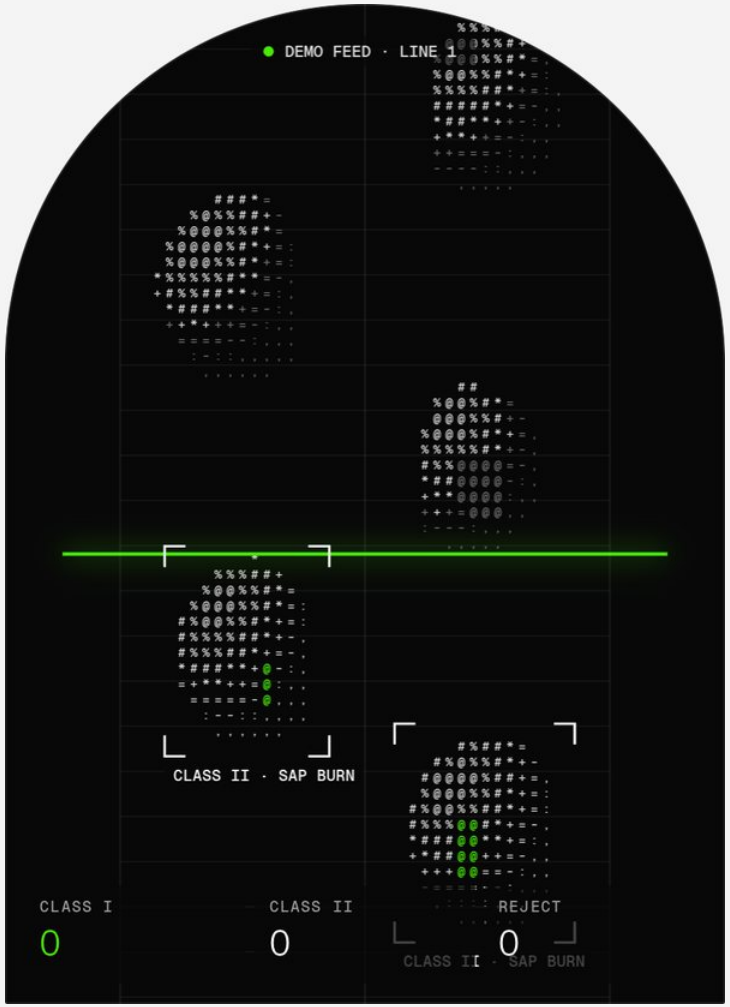
Risk from stage, wetness and temperature; a suggested window.

Photo. Video. Live.

PHOTO	A phone photo of a leaf, a flower panicle or a fruit, answered in the field. Built to work offline first.
VIDEO	Walk a row with a phone or fly a drone. Every tree in frame is found, counted and filed to its block.
LIVE	Fixed cameras on the canopy or the pack line watch quality and pests around the clock.

On the pack line, cameras size each mango and flag blemish, sap burn and rot, so the grade does not depend on who is on shift. One deep-learning system reported 98% on a public mango image set; agc162 will be measured on each client's own line.

FOODS 11(21): 3483, 2022



DEMO PACK LINE, DRAWN IN ASCII

EVIDENCE

The science is ready.

0.968

F1 score for finding mangoes on tree canopies, on an independent test set.

KOIRALA ET AL. 2019

○ MODERATE

4.6–15.2%

Camera fruit-load estimates within this range of packhouse counts, across five orchards.

KOIRALA ET AL. 2019

○ MODERATE

≥18 h

of canopy leaf wetness triggered the best-performing anthracnose sprays in mango trials.

DODD ET AL. 1991

○ LIMITED

25–50%

fewer fungicide sprays when a forecast model set the timing on pear, with similar control.

LLORENTE ET AL. 2002

○ MODERATE

47–73%

less spray used by a sensor-guided orchard sprayer in field trials.

USDA-ARS · OHIO STATE

○ MODERATE

39.6% → 6.2%

mango losses in one Sri Lankan chain, conventional against improved handling.

HERATH ET AL. 2021

○ LIMITED

EVIDENCE LABELS: MODERATE = SEVERAL WELL-DESIGNED STUDIES AGREE · LIMITED = FEW STUDIES, ONE SITE OR A SHORT RUN. NONE OF THESE RESULTS ARE AGC162'S OWN.

PILOT

The proof is in the field.

Every tree

scanned each week, not a sample.

○ TARGET

48 h

of warning before each high-risk infection window.

○ TARGET

–25%

fungicide sprays against the calendar, at equal control.

○ TARGET

Per class

accuracy reported on each client's own blocks.

○ COMMITMENT

TARGETS FOR PILOT VALIDATION, NOT RESULTS

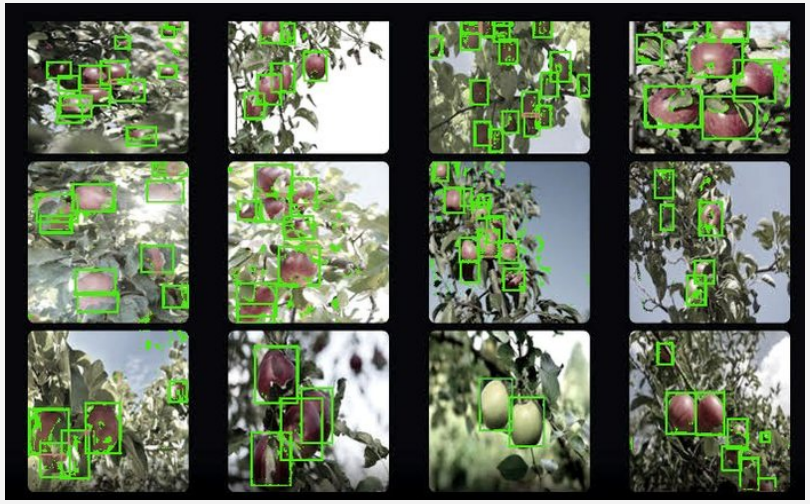
HOW WE WILL MEASURE

Leave-one-orchard-out testing: a model is always scored on an orchard it has not seen.

Precision and recall for every class, per orchard and per season.

Below a confidence threshold, agc162 asks for a closer photo or a scout visit.

Outcomes against a calendar-sprayed control block: sprays, disease at harvest, packout.



FIELD IMAGES LABELLED FRUIT BY FRUIT (ILLUSTRATION)

BUSINESS

Priced per hectare, per season.

MARKET

59 Mt

mangoes, mangosteens and guavas a year; India grows about 44%.

FAOSTAT VIA JIRCAS 2024

$2.8B → $8.5B

AI in agriculture, 2025 to 2030. Another analyst: $2.55B to $7.05B.

BCC RESEARCH 2025 · MORDOR 2025 · COMMERCIAL

PLANS

Scout

Phone scans, per user.

Orchard

Video, drone and spray timing, per hectare per season.

Packhouse

Line cameras, per line.

WHO BUYS · PROPOSED

Orchards

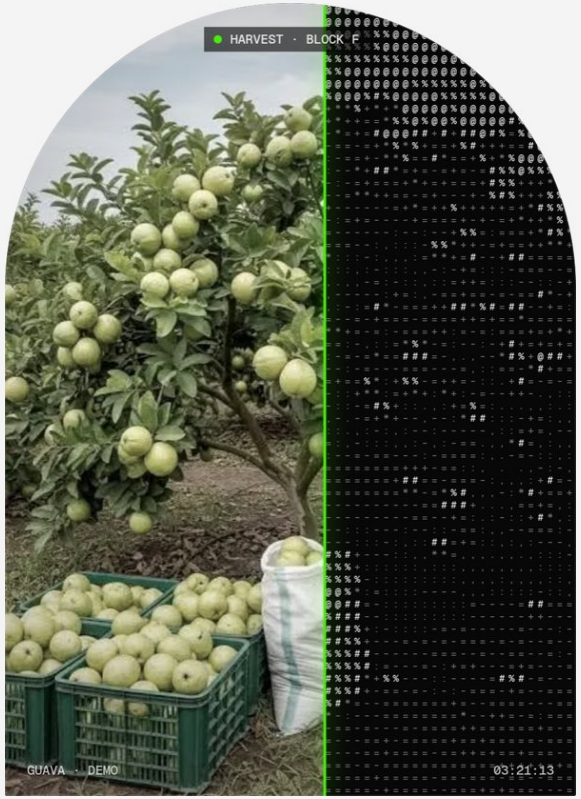
Commercial mango and guava growers.

Packhouses

Export packers who grade and certify.

Advisers

Agronomists and grower groups who scout for many farms.



One season to prove it.

PATH · PROPOSED

Season 1
Mango pilot: phone scans and spray timing, three orchards.

Season 2
Video and drone passes; guava.

Season 3
Pack-line grading.

RISKS WE PLAN FOR

Field accuracy
Scored per orchard; abstains when unsure.

Wrong advice
Suggests registered product classes only; the grower decides.

Adoption
Phone first, offline first, one screen per decision.

THE ASK

Pilot orchards
Three orchards for one flowering-to-harvest season.

Research partners
Plant pathology and post-harvest groups.

Contact
lastfomo@gmail.com

hortus

What grows.

5 DISCIPLINES / 1 HOUSE / HARDWARE BUILT IN-HOUSE

Plant science, farm analysis, crop optimisation, medical plants, and the feeding systems and hardware we build ourselves.

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SOURCES

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RESEARCHED AND DRAFTED WITH AI ASSISTANCE · SOURCES CHECKED OCTOBER 2026 · NUMBERS CARRY THEIR CONTEXT: CROP, PLACE AND STUDY DESIGN